

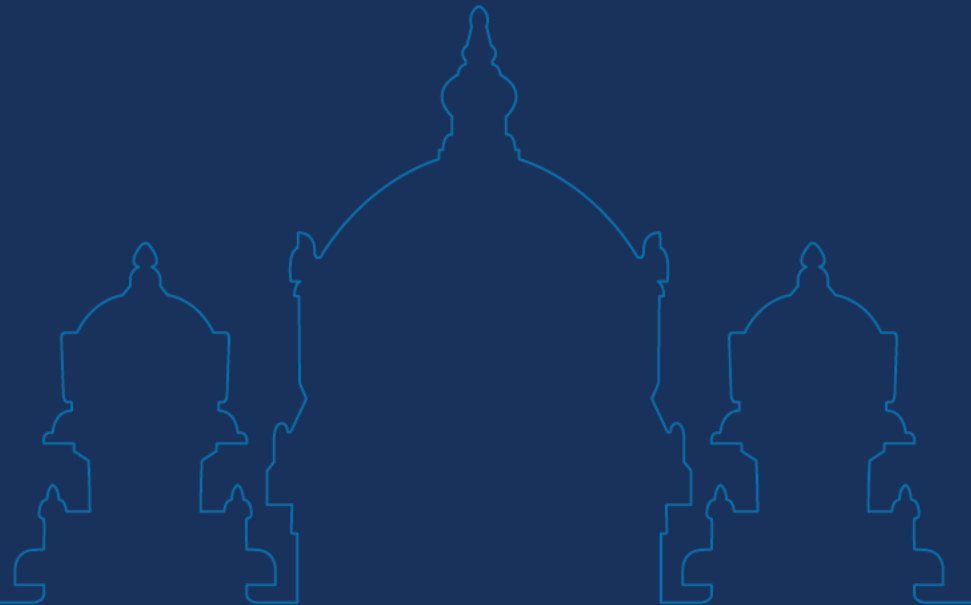
Data Science and Visualization

Unit II DESCRIPTIVE STATISTICS



KCED

Learning Augmented



Unit II – Descriptive Statistics

Advanced Attribute Types - Measures of Central Tendency - Measures of Dispersion - Skewness and Kurtosis - Visual Outlier Identification - Missing Data Visualization - Exploratory Data Analysis (EDA) Techniques - Histograms, KDEs, Box Plots, and Violin Plots.

CO2

Perform Exploratory Data Analysis by applying descriptive statistics and univariate/bivariate visualizations to uncover data distributions, patterns, and relationships, while analyzing data quality through the detection of outliers and exploration of missing value patterns.

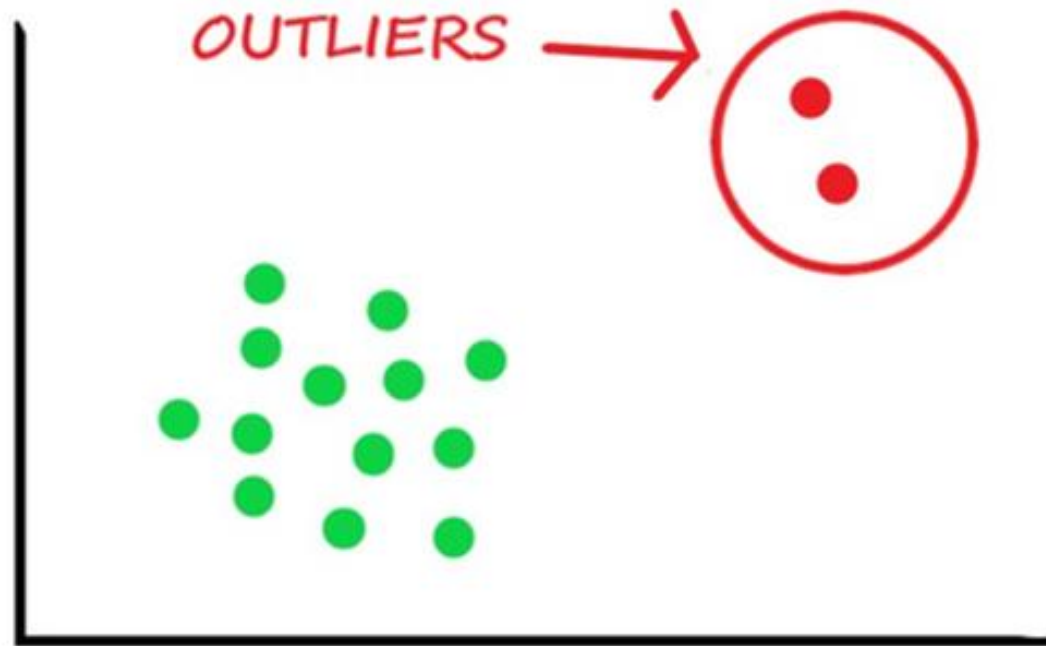
Unit II – Descriptive Statistics

Deep Dive into Data/Attribute Types: Nominal-Ordinal-Binary-Numeric -Discrete vs. Continuous-Asymmetric Attributes-Measurement Scales - Basic Statistical Descriptions: Measures of Central Tendency -Mean-Median-Mode - Measures of Dispersion -Range-Quartiles-Variance-Standard Deviation-IQR - Understanding Skewness and Kurtosis -Conceptual - **Data Cleaning Context: Visual identification of outliers** - Visualizing Missing Data –Types of Missingness- Exploratory Data Analysis Visualization Techniques-Univariate: Histograms with Kernel Density Estimation-Box Plots -Violin Plots.

CO2

Perform Exploratory Data Analysis by applying descriptive statistics and univariate/bivariate visualizations to uncover data distributions, patterns, and relationships, while analyzing data quality through the detection of outliers and exploration of missing value patterns.

Visual Outlier Identification



What are outliers?

- Outliers are **data points that deviate significantly** from the rest of the dataset.
- They can:
 - ✓ Represent **extremely high or low values** compared to others.
 - ✓ The data point(s) which are **significantly different** from other data points.
 - ✓ The data points which are **distributed far away** from the majority data distribution.
- In simple terms, ***outliers are unusual values that don't follow the general pattern of the data.***

Why Do Outliers Occur?

- Outliers are not always bad data.
- **Understanding why they occur helps decide whether to remove, correct, or analyze them.**
 - ✓ Data Entry Errors
 - ✓ Measurement / Unit Errors
 - ✓ Instrument Faults
 - ✓ Sampling Issues
 - ✓ Natural Variability (Meaningful Outliers)
 - ✓ Fraud or Anomalous Behavior

Why Do Outliers Occur?

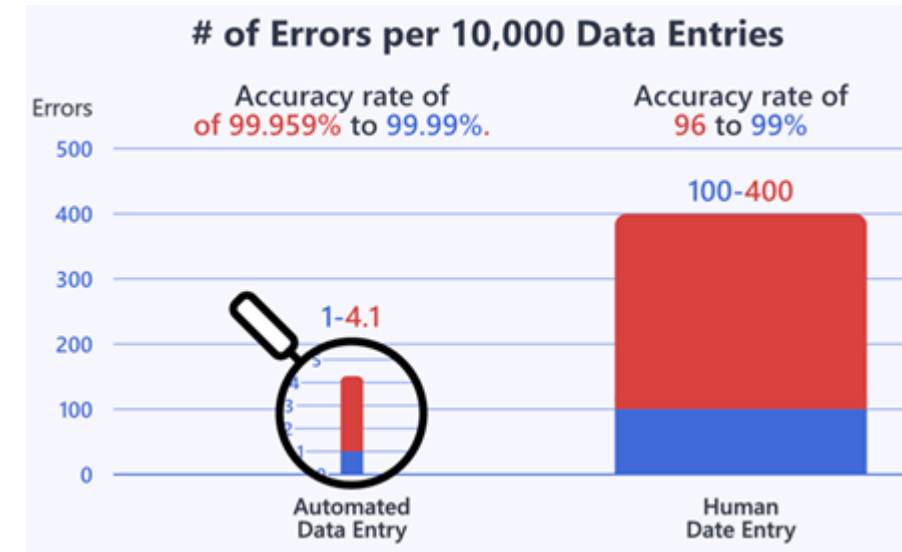
1. Data Entry Errors

- Caused by **manual mistakes or unit mismatches**.
- Create artificial extreme values.

Example:

- ✓ Age entered as 220 instead of 22.
- ✓ Height recorded as 180 m instead of 180 cm.

These outliers should usually be corrected or removed.



Why Do Outliers Occur?

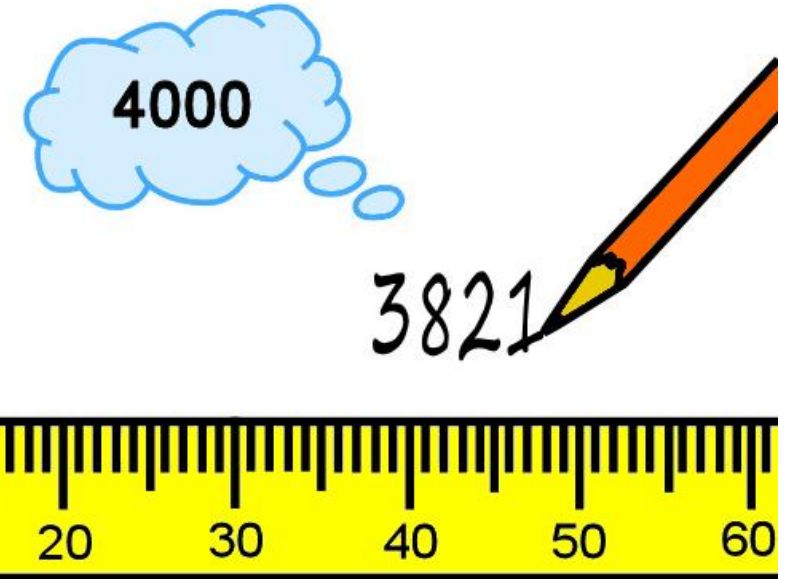
2. Measurement / Unit Errors

- Occur when **different units are mixed** in the same dataset.
- Values look extreme due to **wrong interpretation**.

Example:

- Heights in **cm** mixed with **inches**.
- 68 in → **172.72 cm**, 72 in → **182.88 cm**.

 ***Fix by converting all values to a common unit.***



Why Do Outliers Occur?

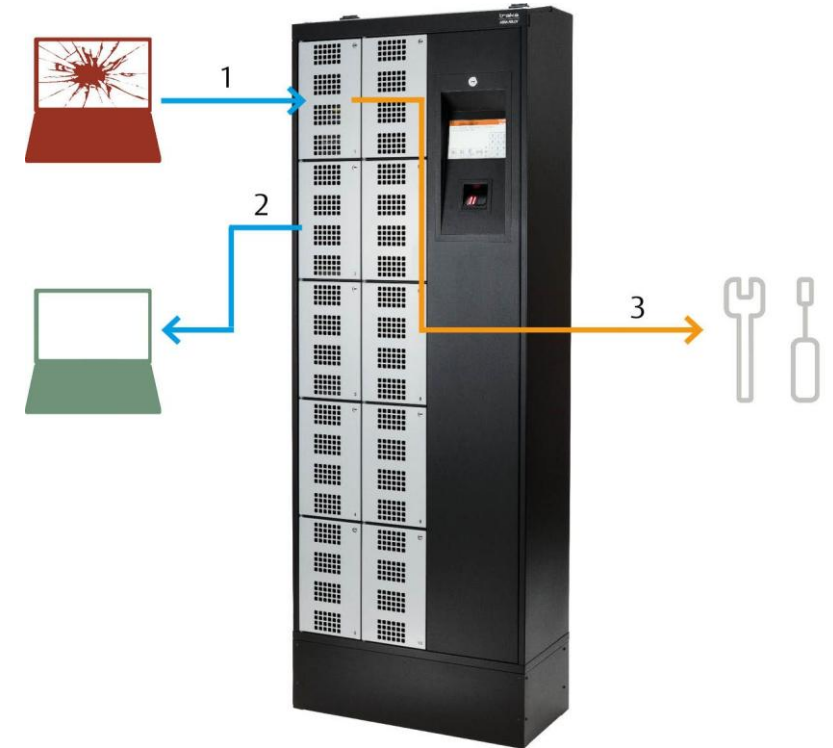
3. Instrument Faults

- Caused by **faulty sensors or devices**.
- Outliers appear due to **incorrect readings**.

Example:

- Temperature sensors show **20–25°C**.
- One sensor reports **80°C** (sunlight exposure / malfunction).

 ***Outlier reflects measurement error, not reality.***



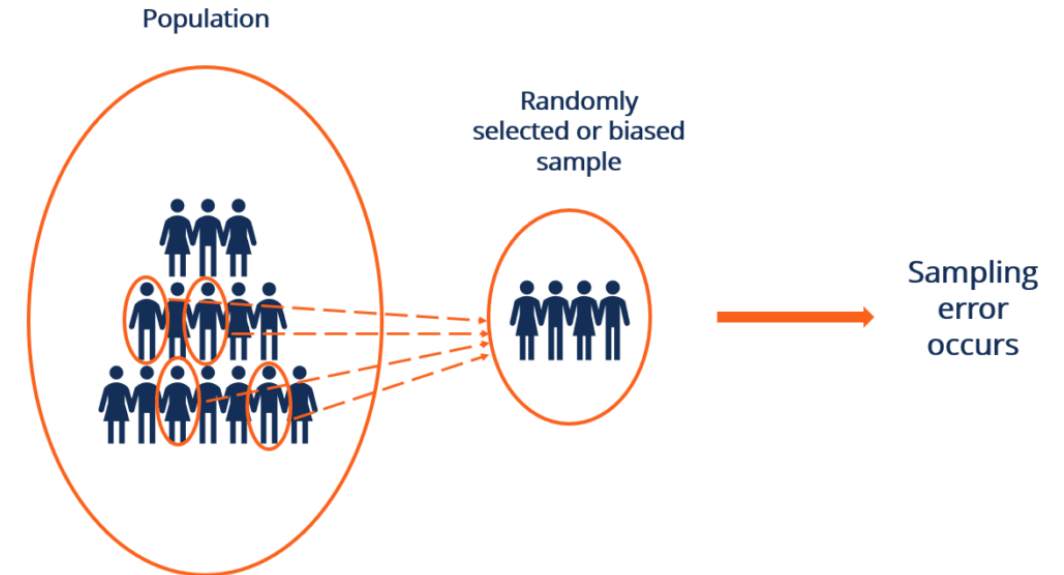
Why Do Outliers Occur?

4. Sampling Issues

- Data combines **different populations**.
- Leads to extreme-looking values.

Example: Mixing college student income with senior executive income.

 ***Segment data before analysis.***



Why Do Outliers Occur?

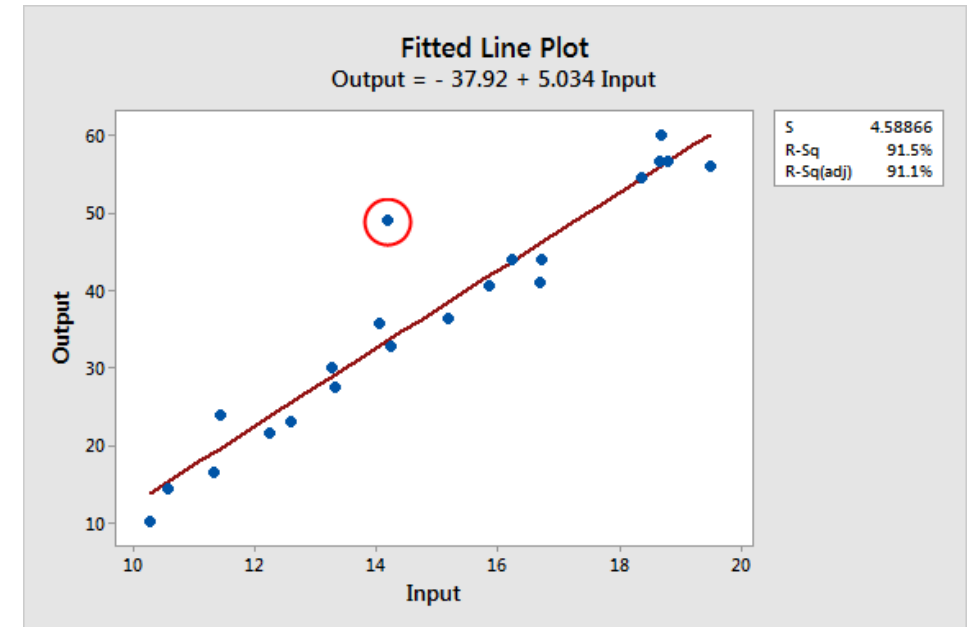
5. Natural Variability (Meaningful Outliers)

- Occur due to **real-world events**.
- Represent **valuable insights**, not errors.

Example:

- Big Billion Day / Diwali Sale.
- Sales jump **5x–10x** (₹12,00,000 in a day).

 ***These outliers show peak demand and successful promotions.***



Why Do Outliers Occur?

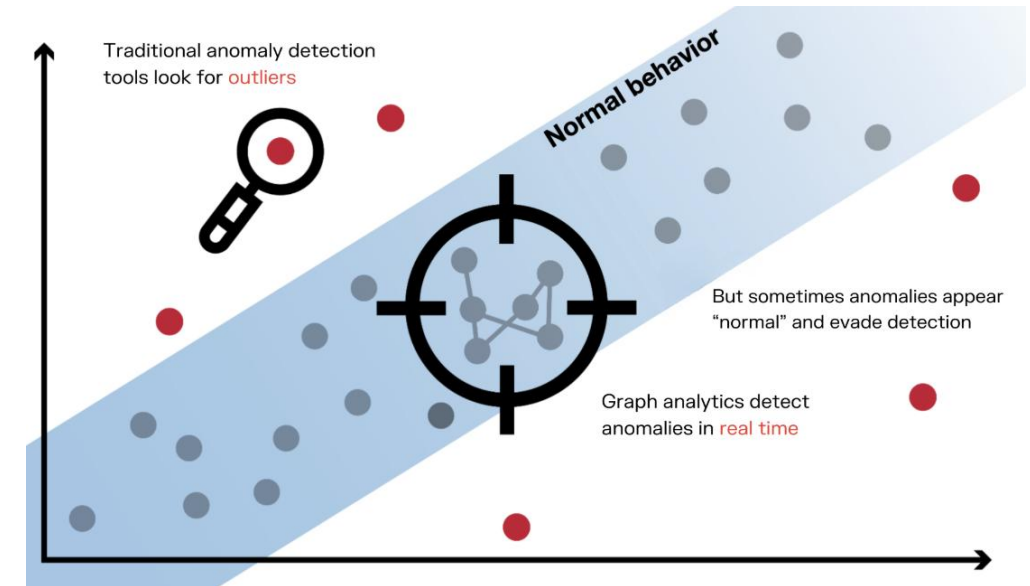
6. Fraud or Anomalous Behavior

- Sudden deviations from normal patterns.
- Often **early warning signs**.

Example:

- Account averages **1,000 transactions/day**.
- Suddenly records **10,000,000 transactions**.

 **Critical outliers → investigate immediately.**



Identify the Reason of Outlier here?

Example: If you have the following dataset of student test scores:

[85, 87, 90, 88, 92, 89, 45]

*The score **45** is an outlier—it's much lower than the others.*

Note: Outliers can be valid observations or errors in data entry, measurement, or processing.



Identify the Reason of Outlier here?

Example: If you have the following dataset of student test scores:

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Data Entry Error / Meaningful Outlier



Impact of Outliers

- Outliers can drastically change the story your data tells.
- Their influence depends on the statistical method or model being used:
 - ✓ Distorted Statistical Measures
 - ✓ Misleading Visualizations
 - ✓ Degraded Machine Learning Model Performance

Impact of Outliers



Statistical Measures



- Distorted Mean & Correlation

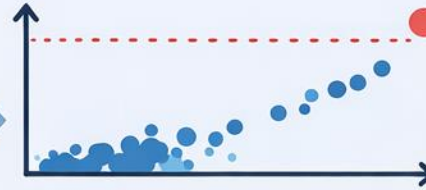


- Distorted Mean & Correlation



Misleading Visualizations

- Axis Stretching, Hidden Patterns



- Axis Stretching, Hidden Patterns



Noise → Outliers



Hurting Machine Learning Models

- Model Bias
- Reduced Accuracy
- Increased Training Time



- ✓ Model Bias
- ✓ Reduced Accuracy
- ✓ Increased Training Time

Noise → Signal

Impact of Outliers

1. Distorted Statistical Measures

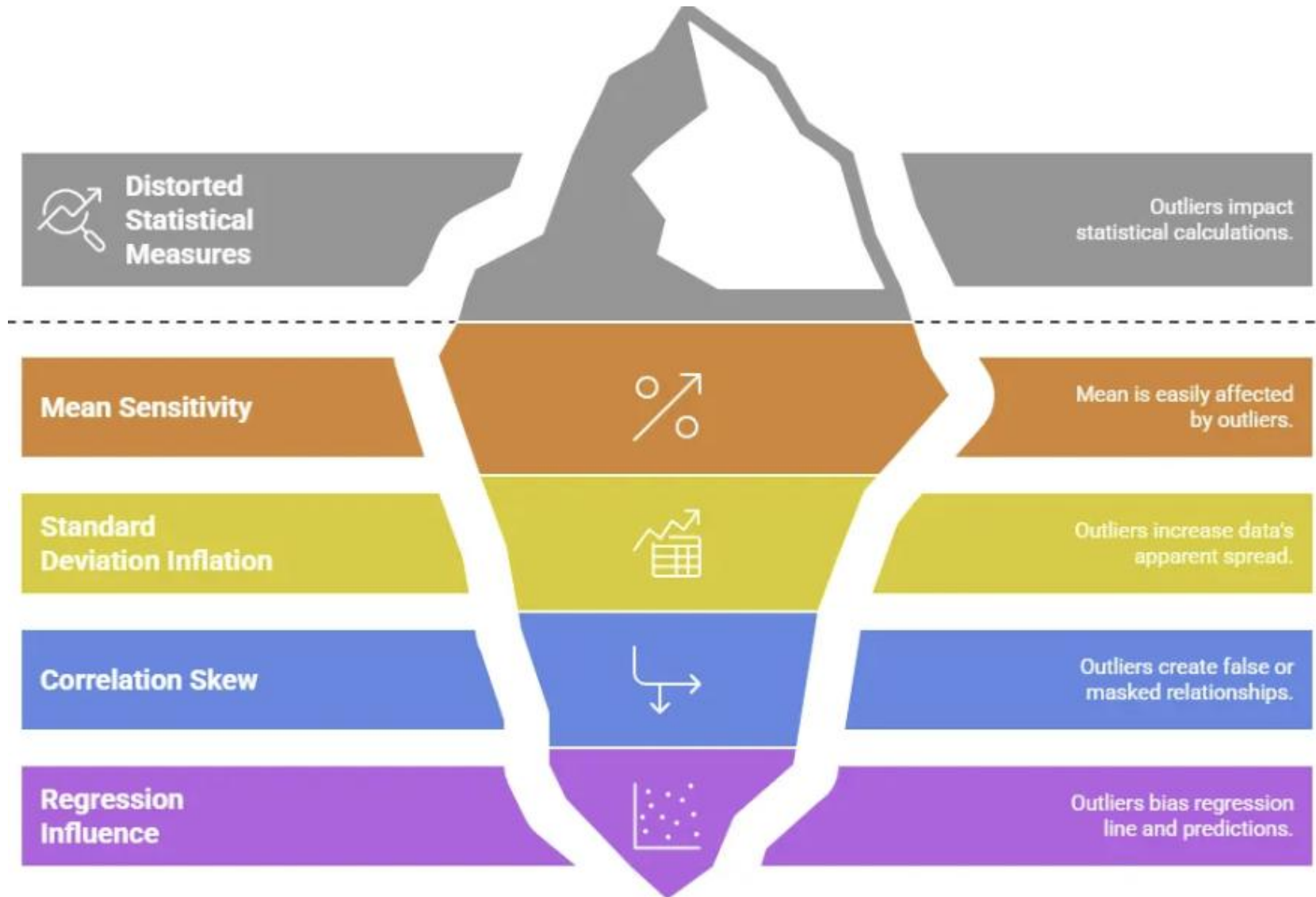
Outliers can **disproportionately affect metrics** like the mean, standard deviation, and correlation.

Example: In a dataset of salaries:

- ✓ Most employees earn between ₹30,000 and ₹70,000.
- ✓ One executive earns ₹10,00,000.

This **single outlier inflates the mean**, making it appear that the **average salary is much higher than it truly is**.

Impact of Outliers

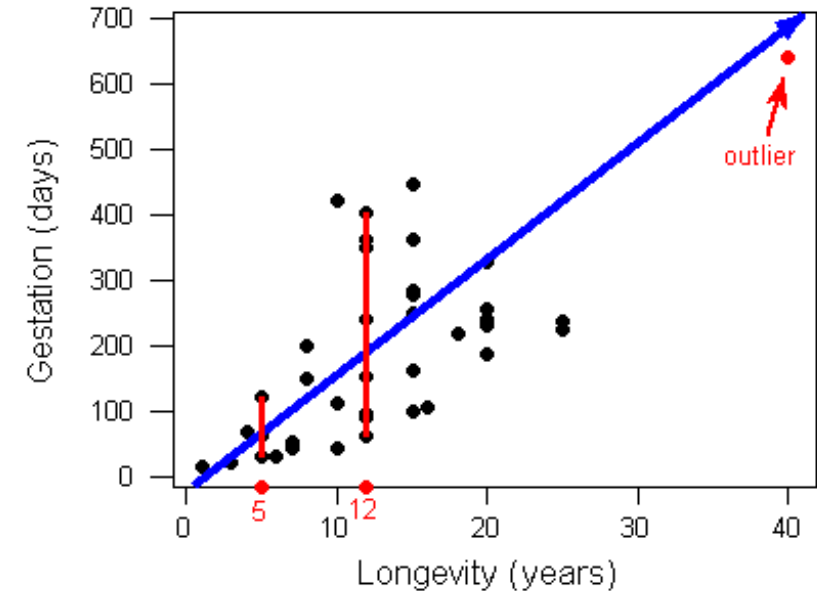


Impact of Outliers

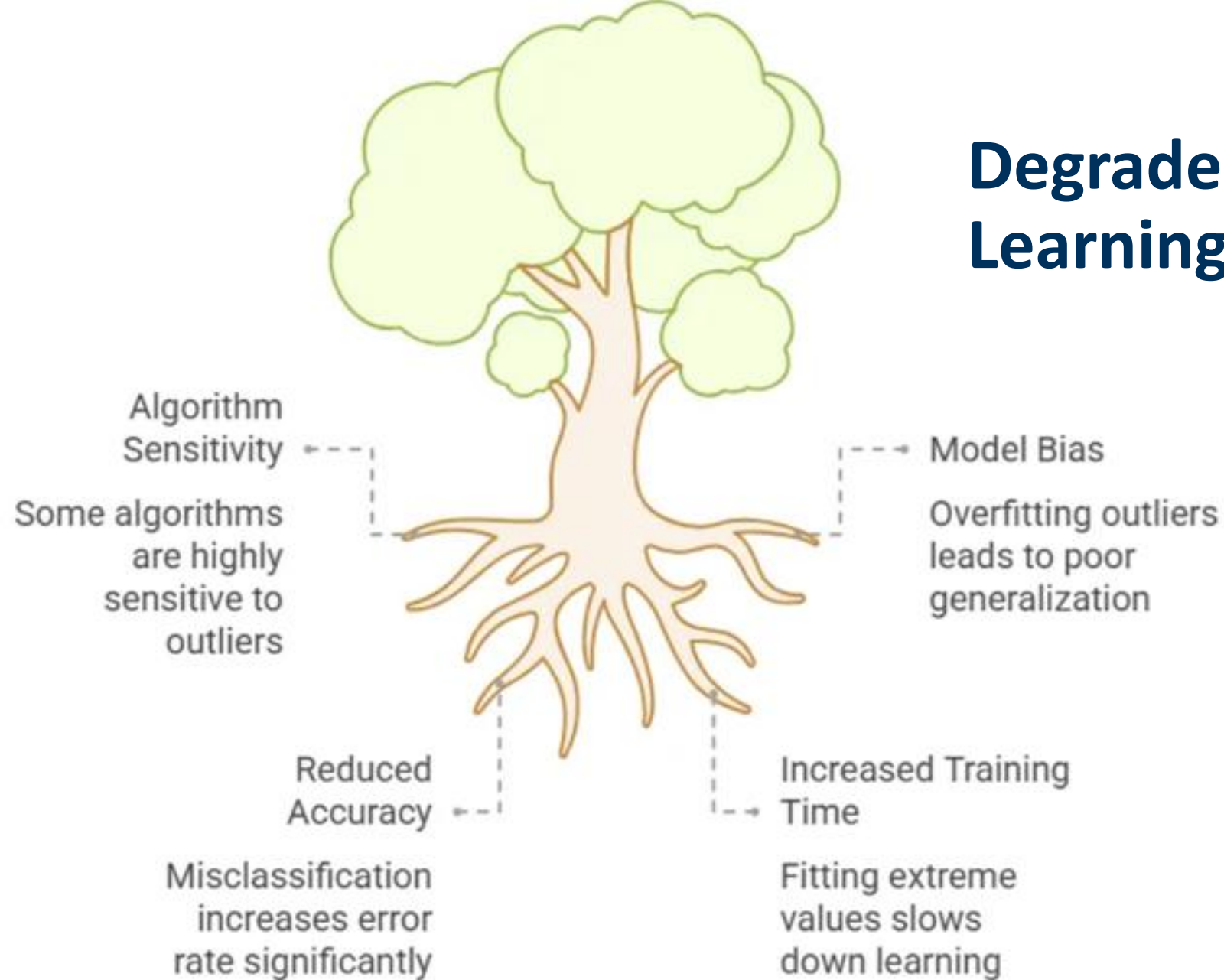
2. Misleading Visualizations

Charts like histograms, boxplots, and scatter plots can become distorted due to outliers, **hiding the true distribution of data**.

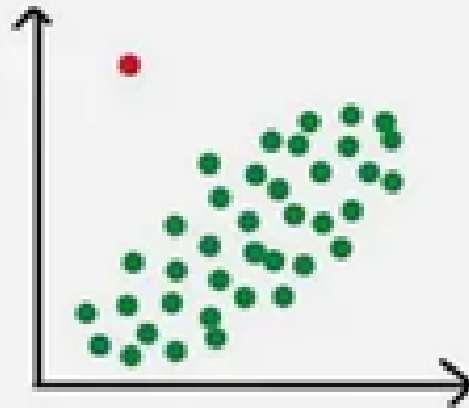
Example: A scatter plot with one extreme value can stretch the axis scale, **compressing the rest of the data into a tiny cluster**.



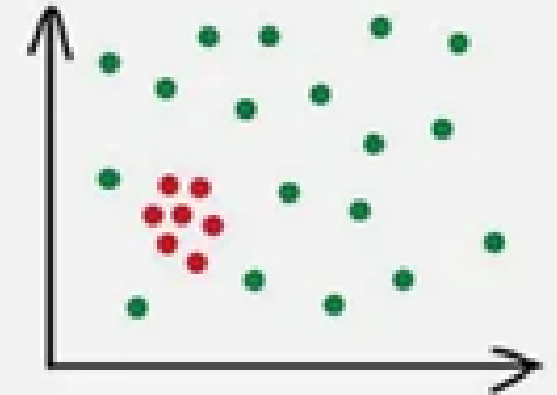
Degraded Machine Learning Performance



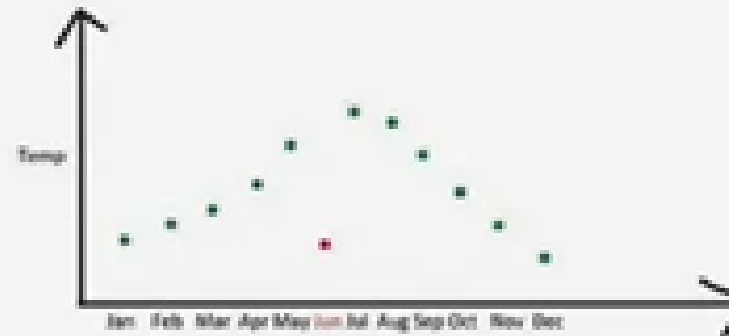
Types of Outliers



Global Outliers



Collective Outliers

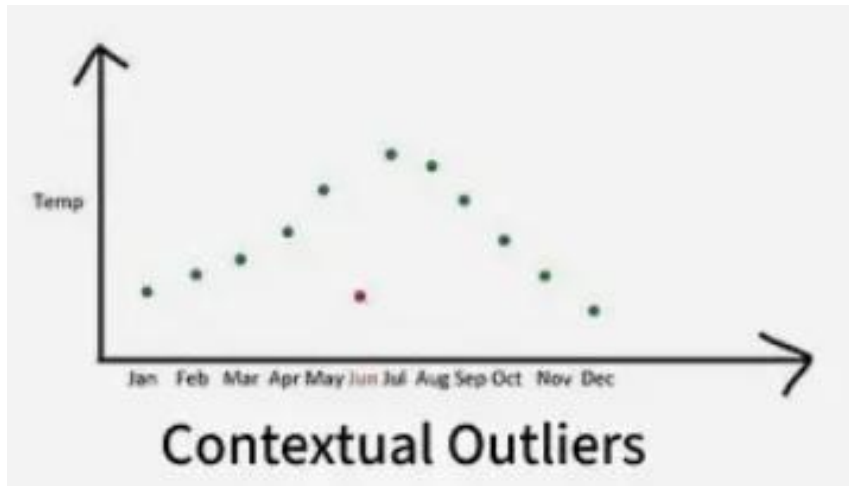


Contextual Outliers

Types of Outliers

1. Global Outliers

Also known as point anomalies, these data points significantly differ from the rest of the dataset. Also known as **Point Outlier**.



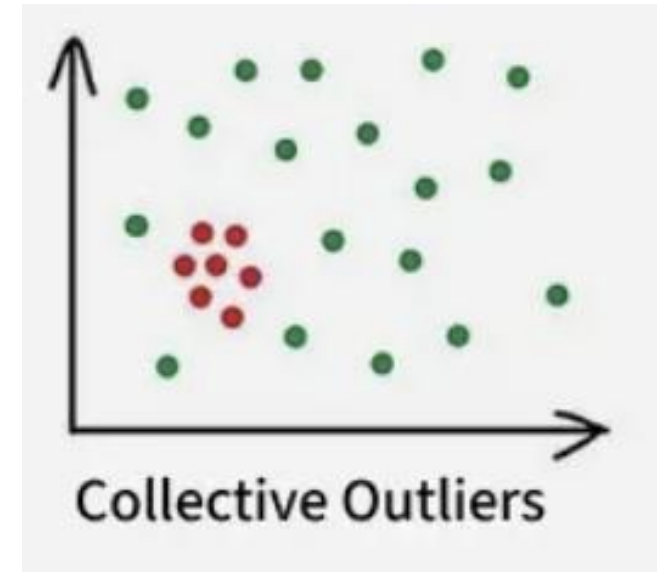
2. Contextual Outliers

These are data points that are considered **outliers in a specific context**. For example, a high temperature may be normal in summer but an outlier in winter.

Types of Outliers

3. Collective Outliers

A **collection of data points** that deviate significantly from the rest of the dataset, even if individual points within the collection are not outliers.



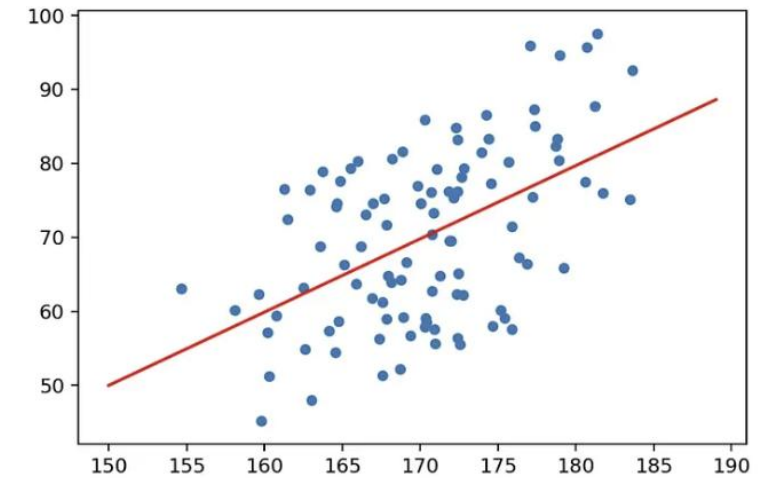
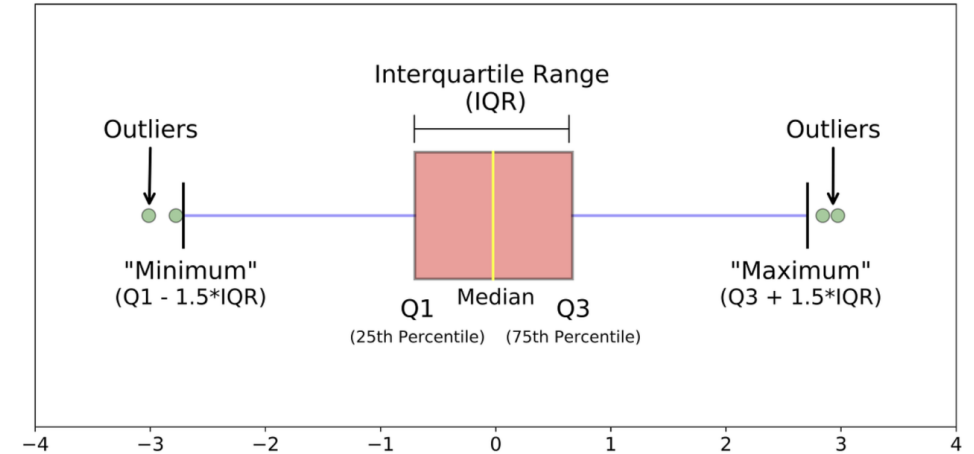
Outlier Detection Methods

	Visualization	▶ Boxplot → Univariate ▶ Scatter Plot → Bivariate
	Statistical Tests	▶ Z-Score Method ▶ IQR Method
	Machine Learning Based	▶ Isolation Forest ▶ K-Nearest Neighbors (KNN) ▶ Local Outlier Factor (LOF)
	Domain Knowledge	▶ Expert-driven thresholds or rules
	Hypothesis Testing	▶ Grubbs' Test

Outlier Detection Methods

1. Visual Inspection:

Box plots, scatter plots, and histograms can help visualize the distribution of the data and identify potential outliers.



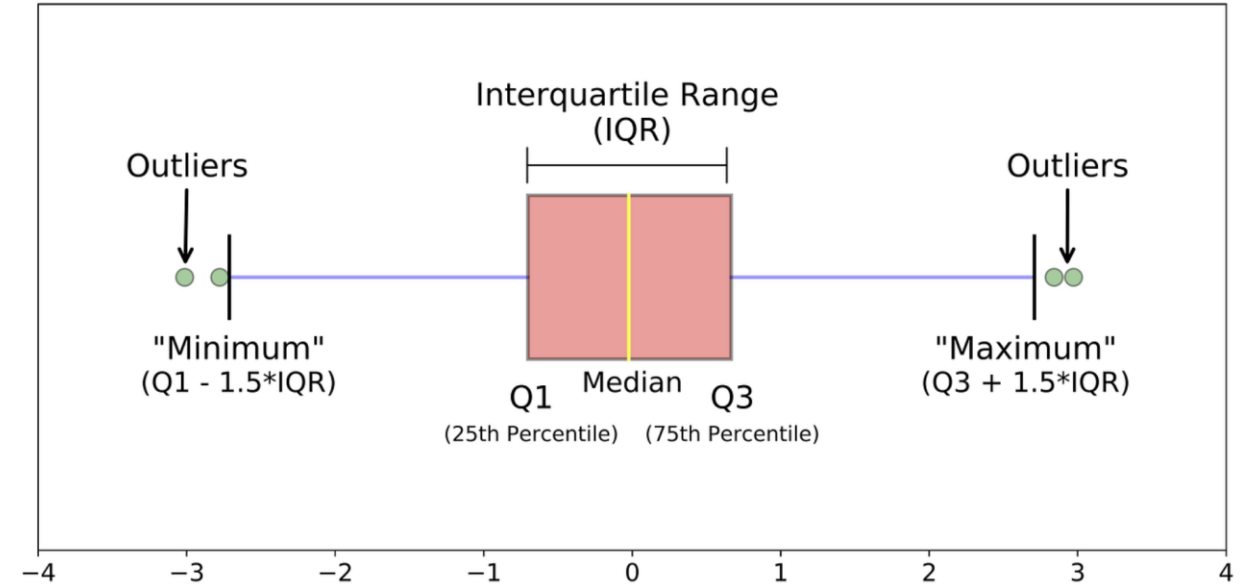
Outlier Detection Methods

Box Plot:

A box plot (box-and-whisker plot) visually shows the distribution of data and helps identify outliers.

Key Statistics

- MinimumQ1
- (First Quartile)
- Median (Q2)
- Q3 (Third Quartile)
- Maximum



How It Works

- ✓ The box represents the **Interquartile Range (IQR)** (middle 50% of data).
- ✓ Whiskers extend up to **$1.5 \times \text{IQR}$** .
- ✓ Points outside the whiskers are **outliers**.

Outlier Detection Methods

Scatter Plot:

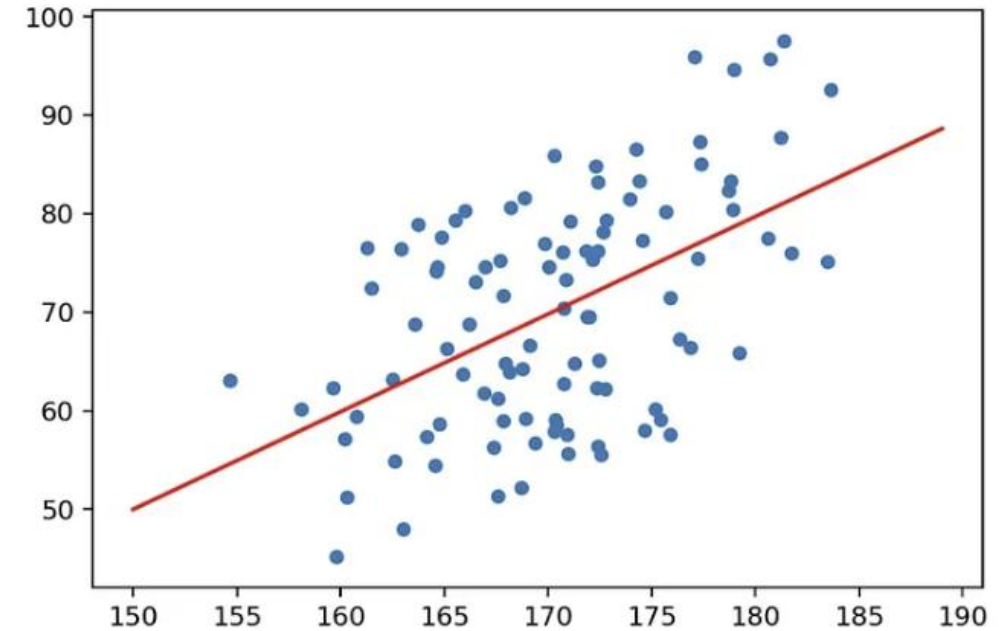
A scatter plot shows the relationship between two numerical variables and helps identify outliers visually.

Correlation Patterns

- Upward trend → Positive correlation
- Downward trend → Negative correlation
- Points close to a line → Strong correlation
- Points widely spread → Weak correlation

How It Works

- ✓ Outliers appear as isolated points.
- ✓ They lie far from the main data cluster.



Outlier Detection Methods

2. Statistical Test:

When analyzing a single variable (or column), we often rely on Z-score and interquartile range (IQR) to detect outliers. These methods are ideal for univariate analysis, where the goal is to identify data points that deviate significantly from the rest of the distribution.

■ Z-Score Method

- Measures distance from the mean in standard deviations
- Common threshold: $Z > \pm 3$
- Best suited for normally distributed data

■ Interquartile Range (IQR) Method

- $IQR = Q3 - Q1$
- Outliers: Below $Q1 - 1.5 \times IQR$ & Above $Q3 + 1.5 \times IQR$
- Effective for skewed or non-normal data

Outlier Detection Methods

3. ML Based:

While statistical methods like Z-score and IQR are typically used for univariate analysis (i.e., analyzing one feature at a time), machine learning-based methods shine when dealing with bivariate or multivariate data, where relationships between multiple features are considered.

- **Isolation Forest:** Detects anomalies by isolating data points. It detects only global outliers.
- **KNN (k-nearest neighbors):** Outliers are points far from their neighbors. It detects only global outliers.
- **Local Outlier Factor (LOF):** Measures local deviation from density of neighbors. It detects both local and global outliers.

Outlier Detection Methods

Algorithm	Detects Global Outliers	Detects Local Outliers	Notes
Isolation Forest	✓ Yes	✗ No	Best for high-dimensional data; isolates anomalies by random splits ¹
K-Nearest Neighbors (KNN)	✓ Yes	✗ No	Distance-based; flags points far from others ²
Local Outlier Factor (LOF)	✓ Yes	✓ Yes	Density-based; compares local density to neighbors ³

Outlier Detection Methods

4. Domain Knowledge:

- Sometimes, statistical methods alone aren't enough to detect outliers. That's where domain knowledge comes in, understanding the rules, norms, and expectations of the data you're working with.
- It allows you to spot values that are technically within range but contextually impossible.

Student	Marks
A	85
B	92
C	150 ← ✖
D	78

Outlier Detection Methods

4. Domain Knowledge:

Example: Exam Marks - Imagine you're analyzing student exam scores. The maximum possible score is **100**. Most students score between 0 and 100. Suddenly, you see a record showing a score of **150**.

Statistically, 150 might not be flagged as an outlier if the distribution is wide enough. But domain knowledge tells you that 150 is not valid; it exceeds the maximum possible score. **This is likely a data entry error.**

5. Hypothesis Testing:

Grubbs' Test: A statistical test to check whether the maximum or minimum value is an outlier in normally distributed data.

Handling Outliers: What to Do After Detection

- Once outliers are identified, the next step is deciding how to handle them.
- Importantly, **not all outliers should be removed**.
- Some indicate errors, while others represent rare but meaningful events.
- The correct treatment depends on:
 - ✓ The **cause** of the outlier
 - ✓ The **domain context**
 - ✓ The **goal** of the analysis or model

Outlier Handling Techniques

- **Remove** – Delete outliers caused by errors or impossible values.
- **Impute / Replace** – Substitute outliers with mean, median, or domain values.
- **Cap / Winsorize** – Limit extreme values to predefined thresholds.
- **Binning** – Group continuous values into intervals to reduce noise.
- **Keep** – Retain outliers when they represent valid rare events.

Method	Description	When to Use	Pros	Cons
Remove	Delete rows containing outliers	Clear errors, invalid values, or extreme distortions	Simple and effective	May lose valuable or rare data
Impute / Replace	Substitute outliers with mean, median, or domain-specific values	Small datasets, continuity needed, or domain rules apply	Preserves row count	Can introduce bias or mask variability
Cap / Winsorize	Limit extreme values to a fixed percentile range	When you want to reduce distortion without deleting data	Retains all data points	May hide true extremes
Binning	Group values into intervals and smooth within bins	When discretization or smoothing is needed	Simplifies data, reduces noise	Loss of detail and granularity
Keep	Retain outliers as-is	When outliers are meaningful (e.g., fraud, rare events, breakthroughs)	Preserves important signals	May skew analysis if not handled properly

Challenges with Outlier Detection

Detecting outliers effectively poses several challenges:

- **Determining the Threshold:** Deciding the correct threshold that accurately separates outliers from normal data is critical and difficult.
- **Distinguishing Noise from Outliers:** In datasets with high variability or noise, it can be particularly challenging to differentiate between noise and actual outliers.
- **Balancing Sensitivity:** An overly aggressive approach to detecting outliers might eliminate valid data, reducing the richness of the dataset.

Applications of Outlier Detection

- **Financial Fraud Detection:** Identifies unusual transactions (e.g., high-value foreign purchase on a normally local card).
- **Cybersecurity:** Detects abnormal network activity indicating intrusions or data leaks.
- **AI / ML Modeling:** Used for data cleaning, preventing model skew, and detecting biased predictions
- **Big Data & Cloud Systems:** Flags unauthorized access and corrupted or anomalous data entries

References

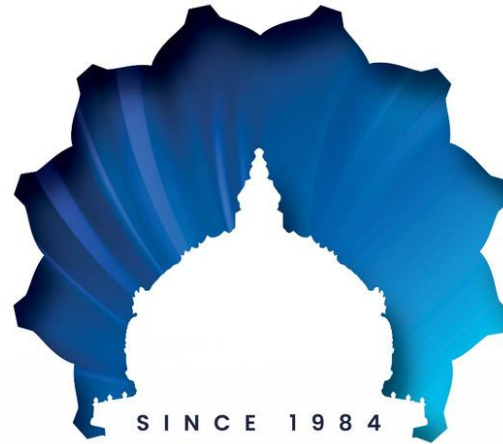
<https://www.geeksforgeeks.org/data-analysis/what-is-outlier-detection/>

<https://www.geeksforgeeks.org/data-analysis/types-of-outliers-in-data-mining/>

<https://medium.com/@anand.bharadwaj9794/data-cleaning-handling-outliers-098b437d4648>

<https://www.numberanalytics.com/blog/7-essential-outlier-analysis-techniques>

<https://www.analyticsvidhya.com/blog/2022/10/outliers-detection-using-iqr-z-score-lof-and-dbscan/>



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